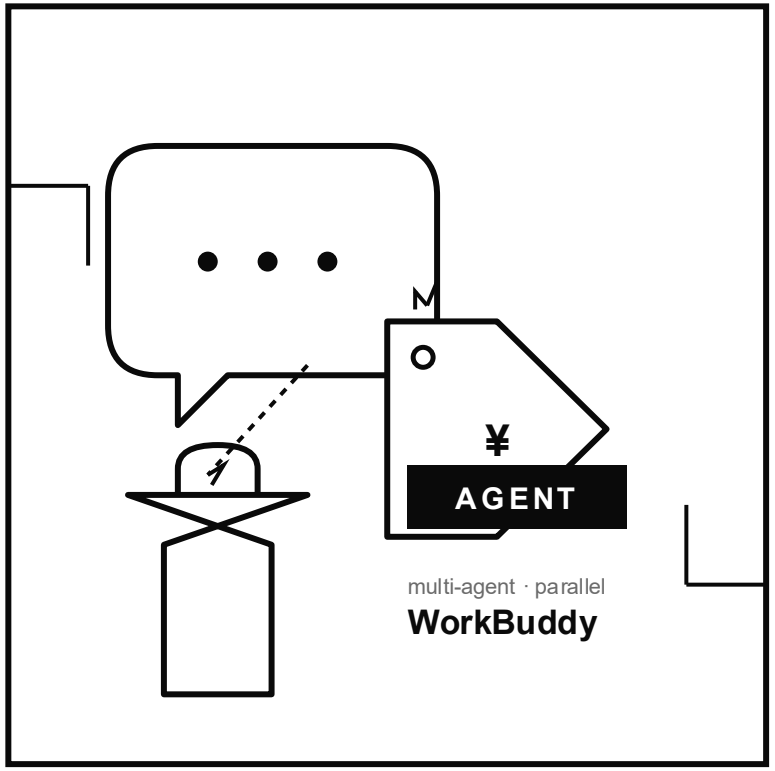
TRACK 2 • WORKBUDDY TRACK (AGENTIC SOLUTIONS)

Campus Second-Hand Trading Agent

One sentence. Find a reliable seller.

Auto-send a WeChat message — end to end.



GROUP • OFF-ROAD ▪ Pitch · 2026

LI XUAN PEI YILAN
ZHOU MO DING ZIHAN

The messy reality of campus second-hand trading.

EVERY CAMPUS HAS ONE OF THOSE WECHAT GROUPS.

Thirty-one demo messages. Jokes, requests, "anyone selling X?", and noise. Hidden inside: genuine listings — and genuine risks.

PAIN 01 **High signal-to-noise ratio**

Real listings buried under chatter. You scroll for 10 minutes to find one usable item.

PAIN 02 **Trust gap**

No verification. Are prices real? Will the seller show up? Is the contact legit?

PAIN 03 **Manual friction**

Ask the seller, copy the contact, draft a message, send it. Every step is on you.

OUR ANSWER

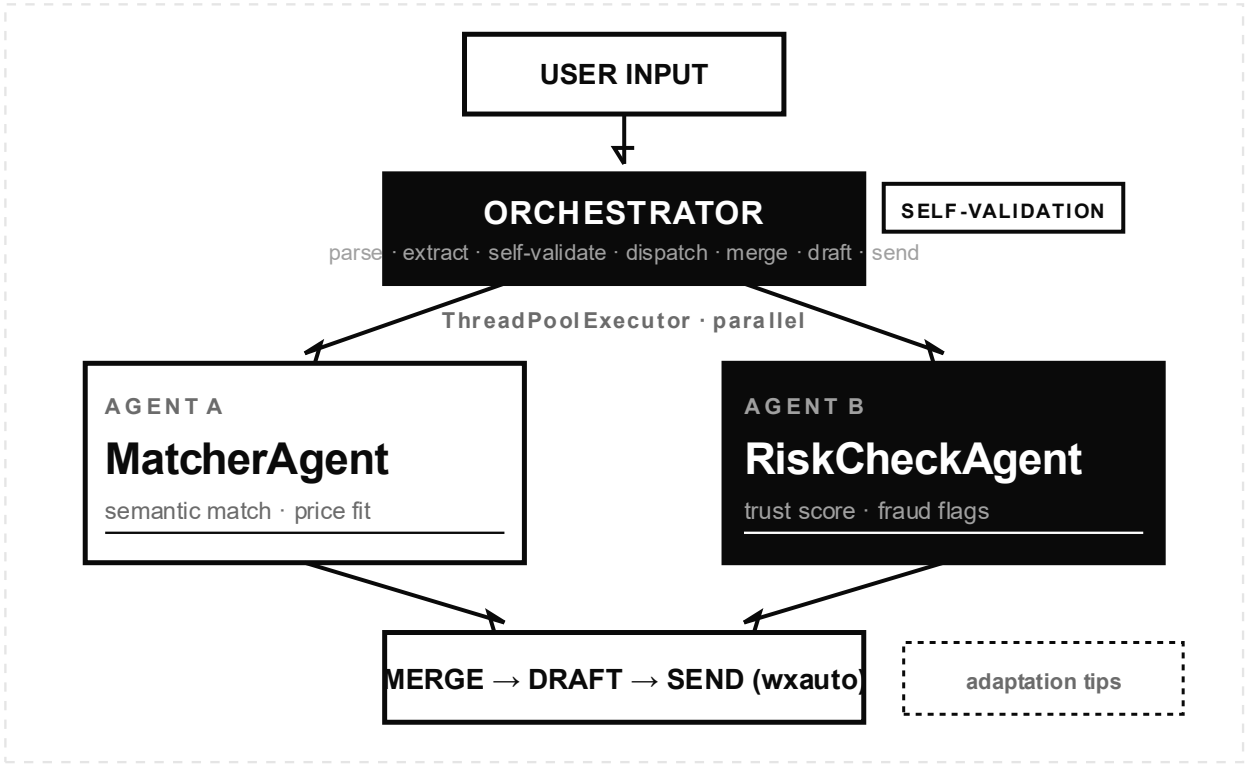
You ask. A multi-agent system does the rest — Orchestrator dispatches Matcher and RiskCheck in parallel, drafts a safe message, and sends it.

31 → 1

demo messages · sentence to deal

One Orchestrator. Two agents. Parallel execution.

MatcherAgent and RiskCheckAgent run concurrently via ThreadPoolExecutor — halving latency, doubling coverage.



KEY DESIGN DECISIONS

PARALLEL	Two agents run concurrently — Matcher scores listings while RiskCheck evaluates trust, cutting latency in half.
SELF-VALIDATION	Orchestrator validates extracted listings before dispatch. Empty results trigger adaptation suggestions, not crashes.
ADAPTATION	Dynamic suggestions guide users when no match is found — "try widening your budget" or "check another group."
RETRY/BACKOFF	LLM calls wrapped in exponential backoff. Every inference site is fault-tolerant by default.

■ **Agent Work Panel — live agent activity in UI**

WorkBuddy is not a wrapper. It is the engine.

Four integration layers. Every LLM call, every parallel agent, every tool action runs through WorkBuddy.

INTEGRATION STACK • 4 LAYERS

NOT A CHAT WRAPPER

01

CONVERSATIONAL INTERFACE

Users type natural language. WorkBuddy parses intent, manages multi-turn context, and surfaces results in the Agent Work Panel.

02

MULTI-AGENT ORCHESTRATION

OrchestratorAgent decomposes tasks and dispatches Matcher + RiskCheck in parallel via ThreadPoolExecutor. Self-validation gates each phase.

03

LLM INFERENCE ENGINE

4+ LLM call sites — extract, match, assess, draft. All routed through WorkBuddy with DeepSeek backend and exponential backoff retry.

04

TOOL ACTION LAYER

WorkBuddy triggers real external actions — wxauto dispatches WeChat messages, file parsing imports chat logs. Not mocked. Not simulated.

PROOF POINTS

3

autonomous agents — Orchestrator, Matcher, RiskCheck — all powered by WorkBuddy's inference pipeline.

- **Agent Work Panel**
  Live UI: agent phases, dispatch, merge status.
- **Parallel execution**
  Matcher + RiskCheck run concurrently — 2x faster.
- **Self-validation**
  Validates results before dispatch. No empty verdicts.
- **Retry & backoff**
  Exponential backoff on every LLM call site.

Integration Depth

TRACK 2 • WORKBUDDY TRACK

Thank You.

Questions are welcome.

ACKNOWLEDGMENTS

To the WorkBuddy team for the execution engine —
and to every campus second-hand group that inspired this build.

■ GROUP • OFF-ROAD

github.com/Leokm123/Campus-Second-Hand-Trading-Agent